Apache Module

Apache is a venerable web server that has been around for decades, and can integrate PHP in the same way as Nginx – as a CGI binary. PHP can also run as an Apache module, where the PHP binary is linked directly to the Apache process (as a `.so` or `.dll` library). The mode under which Apache is running PHP doesn't functionally impact the language: from PHP's perspective one web server is pretty much as good as another. Where this book touches on aspects of the language that need a web server environment we'll almost always use Nginx or PHP's `cli-server` SAPI, but I'll toss at least one Apache-driven example into the book somewhere to keep things (even more) interesting. If you're interested in learning about Apache, start here: `https://httpd.apache.org/`

Configuring PHP

PHP has many configurable options that control its behavior during execution. There are settings for logging errors (where to write, how much to write, and what types of errors), maximum memory use, controlling session data (for web-server SAPIs), and more. These options are collectively known as *directives* and are usually held in special files called *ini files* (named for their filename extension). Directives control both core PHP settings and those for extensions, and generally have default values when they are not explicitly set.

System-wide directives are typically located alongside other system-level configuration files such as the `/etc` path on Linux systems, or within the Apache config files (if running as an Apache module) such as `httpd.conf`. We'll look closely at this later on in this section.

Directives can also be set on a per-directory basis: every script in the same directory (or a descendant sub-directory) as the config is affected. This varies depending on the platform. For the CGI/FastCGI SAPI the files are named `.user.ini` – if running as an Apache module (`mod_php`) they are `.htaccess` files. If PHP is running on a Windows platform it also possible for some directives to be set via the system registry.

Finally, certain directives can be set during script execution using the built-in `ini_set()` function. It is also possible for PHP binaries (CLI, CGI/FastCGI SAPIs) to be invoked with directives as command line options.